UNIT 5 · 9 hours

Op-Amp as a Comparator



Op-Amp and Linear IC Applications

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

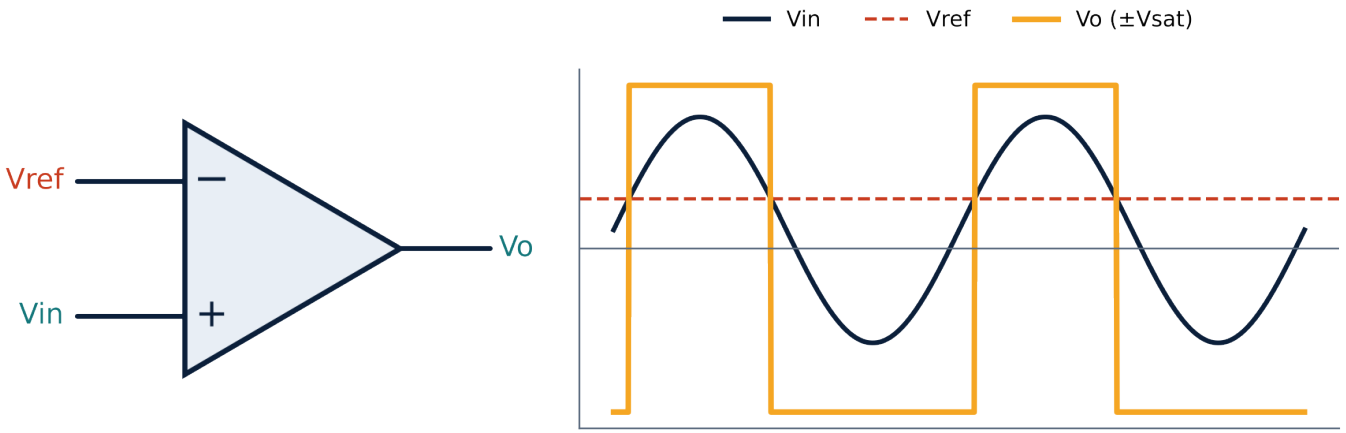
- Explain the working of an op-amp comparator
- Draw output waveforms for sine input and fixed reference
- Describe a zero-crossing detector
- Mention hysteresis (Schmitt trigger) and applications

Syllabus context

Introduction, op-amp specifications, differential and common-mode operation, op-amp applications: inverting/non-inverting amplifier, summing, integrator, differentiator, comparator, 555 timer IC in astable mode, 78XX series IC voltage regulators.

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Comparator operation



- Op-amp used in open loop (no feedback).
- Vin > Vref: output saturates at +Vsat.
- Vin < Vref: output at −Vsat.
- Compares two voltages and gives a digital-like output.

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Non-inverting comparator: Vin at +, Vref at −: output high when Vin > Vref.

Zero-crossing detector: Vref = 0 V; the output changes sign at each zero crossing of the input.

Output levels: about ±(VCC − 1.5 V), or 0 and +V with a single supply.

Limitations: noise near the threshold causes chatter; response limited by slew rate.

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Schmitt trigger (comparator with hysteresis)

Positive feedback

A fraction of the output is fed back to the + input.

Two thresholds

Upper VUT and lower VLT: the output switches at different levels for rising and falling input.

Benefit

Immune to noise near the threshold; cleans slow or noisy signals.

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Numerical: comparator timing

Problem Vref = 2 V, vin = 5 sin(100πt) V, saturation ±13 V. Find the fraction of the cycle for which the output is high and the high time.

- vin = 2 V at sin θ = 0.4 → θ1 = 23.6°, θ2 = 180° − 23.6° = 156.4°
- High for 156.4° − 23.6° = 132.8° of 360°
- Fraction = 132.8/360 = 0.369; T = 20 ms

Answer: Output high for 36.9 % of the cycle = 7.4 ms

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Applications

- Zero-crossing and level detectors.
- Analog-to-digital converters (flash and successive approximation).
- Square-wave generation from sine waves.
- Over-voltage / temperature alarms.

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Key takeaways

- Comparator: open-loop op-amp output = ±Vsat
- Vin > Vref: +Vsat; Vin < Vref: −Vsat
- Zero-crossing detector: Vref = 0
- Schmitt trigger adds hysteresis

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Practice questions

Q1. Explain how an op-amp works as a comparator.

Q2. Vref = 1 V, vin = 3 sin ωt. Sketch the output.

Q3. Why is hysteresis needed in a noisy environment?

Q4. Name two applications of comparators.